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Computational Logic Handouts

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Chapter 1

LoP

1.1 Well Formed Formulas

Exercise 1.1 Which of the following symbols are used in LOP?

- | | | |
|-------------|------------------|----------------|
| 1. $\sqcap$ | 6. $\sqsubseteq$ | 11. $\oplus$ |
| 2. $\neg$ | 7. $\supset$ | 12. $\forall$ |
| 3. $\vee$ | 8. $\wedge$ | 13. $\diamond$ |
| 4. $\equiv$ | 9. $\models$ | 14. $\square$ |
| 5. $\sqcup$ | 10. $\exists$ | 15. $\otimes$ |

Exercise 1.2 Which of the following formulas are NOT syntactically correct in LOP?

1. $\neg \text{MonkeyLow} \vee \text{BananaHigh}$
2. $\neg \neg \text{MonkeyLow} \neg \text{BananaHigh}$
3. $\text{MonkeyLow} \neg \neg \text{BananaHigh}$
4. $\text{MonkeyLow} \supset \neg \text{GetBanana}$
5. $\text{MonkeyLow} \equiv \text{BananaHigh}$
6. $\neg (\text{MonkeyLow} \vee \text{BananaHigh}) \supset \text{BananaHigh}$

Exercise 1.3 Which of the following are well formed propositional formulas?

1. $\vee pq$
2. $(\neg(p \supset (q \wedge p)))$
3. $(\neg(p \supset (q = p)))$
4. $(\neg(\diamond(q \vee p)))$
5. $(p \wedge \neg q) \vee (q \supset r)$
6. $p \neg r$

1.2 Models and Theories

Exercise 1.4 Let A and B be propositions. Let $T1 = \{(A \supset B)\}$ and $T2 = \{\neg A \wedge B\}$. Say which of the following sentences is true.

1. $T1$ has 2 models.
2. $T2$ has 1 model.
3. $T1$ and $T2$ have 2 models in common.
4. Does $T1 \models T2$?
5. Does $T2 \models T1$?
6. $T1$ and $T2$ are maximal theories for $M = \{B\}$.

Exercise 1.5 Let A and B be propositions. Let $T1 = \{(A \equiv B)\}$ and $T2 = \{\neg A, B\}$. Say which of the following sentences is true.

1. $T1$ has 2 models.

2. T_2 has 1 model.
3. T_1 and T_2 have 1 model in common.
4. Does $T_1 \models T_2$?
5. Does $T_2 \models T_1$?
6. T_2 is a maximal theory for $M = \{B\}$.
7. T_1 is a maximal theory for $M = \{A, B\}$.
8. The minimal model of T_2 is $M = \{B\}$.
9. The minimal model of T_1 is $M = \{A, B\}$.

Exercise 1.6 RECALL: A minimal Model M of T is the intersection of all models of T .

Let A, B, C be propositions. Given the theory $T = \{\neg A \vee \neg B, C \wedge \neg B\}$, say which of the following sentences is true.

1. T has 2 models.
2. T has 1 model.
3. There is no minimal model of T .
4. T satisfies $M = \{A\}$.
5. The minimal model is $M = \{C\}$.
6. The minimal model is $M = \{A, C\}$.

Exercise 1.7 RECALL: A minimal Model M of T is the intersection of all models of T .

Let A, B, C be propositions. Given the theory $T = \{A \wedge \neg B, A \wedge C, C \wedge \neg B\}$, say which of the following sentences is true.

1. T has 2 models.
2. T has 1 model.
3. There is no minimal model of T .
4. T satisfies $M = \{A\}$.
5. The minimal model for T is $M = \{A\}$.
6. The minimal model for T is $M = \{A, C\}$.

Exercise 1.8 Given propositions X, Y and Z and two theories $T_1 = \{\neg(X \equiv Y), Y \wedge Z\}$ and $T_2 = \{\neg X, Y, Z \supset Y\}$ in $\mathcal{LOP}$ language, indicate which of the following statements are true (one or more).

1. T_1 has 2 models
2. T_2 has 2 models
3. T_1 and T_2 have 1 model in common
4. $T_1 \models T_2$
5. $T_2 \models T_1$
6. $M = \{Y, Z\}$ is a model for T_1
7. $M = \{X, Y\}$ is a model for T_2
8. The minimal model of T_2 exists and is $M = \{Y\}$

Exercise 1.9 Say which of the following statements are true:

1. If a fact is an element of a $\mathcal{LODE}$ domain then the proposition representing it in the $\mathcal{LOP}$ language describing the domain is true.
2. If a proposition in a $\mathcal{LOP}$ language is true then the fact represented in the $\mathcal{LODE}$ domain described by the $\mathcal{LOP}$ language is true.
3. Given an $\mathcal{EG}$ formalized in a $\mathcal{LODE}$ logic, for each assertion in the $\mathcal{LODE}$ language there can exist multiple (atomic) propositions in the corresponding theory formalized in $\mathcal{LOP}$.
4. An Entity Graph (ABOX) formalized in $\mathcal{LODE}$ logic can have multiple models.

1.3 I2F

Exercise 1.10 Let's consider a propositional language where

- p means *Paola is happy*,

- q means *Paola paints a picture*,
- r means *Renzo is happy*.

Formalize the following sentences:

1. *if Paola is happy and paints a picture then Renzo isn't happy*
2. *if Paola is happy, then she paints a picture*
3. *Paola is happy only if she paints a picture*

Exercise 1.11 Let's consider a propositional language where

- p means *x is a prime number*,
- q means *x is odd*.

Formalize the following sentences:

1. *x being prime is a sufficient condition for x being odd*
2. *x being odd is a necessary condition for x being prime*

Exercise 1.12 Let's consider a propositional language where

- A means *Aldo is Italian*
- B means *Bob is English*

Formalize the following sentences:

1. *Aldo isn't Italian*
2. *Aldo is Italian while Bob is English*
3. *If Aldo is Italian then Bob is not English*
4. *Aldo is Italian or if Aldo isn't Italian then Bob is English*
5. *Either Aldo is Italian and Bob is English, or neither Aldo is Italian nor Bob is English*

Exercise 1.13 Let's consider a propositional language where

- A means *Angelo comes to the party*,
- B means *Bruno comes to the party*,
- C means *Carlo comes to the party*,
- D means *Davide comes to the party*.

Formalize the following sentences:

1. *If Davide comes to the party then Bruno and Carlo come too*
2. *Carlo comes to the party only if Angelo and Bruno do not come*
3. *Davide comes to the party if and only if Carlo comes and Angelo doesn't come*
4. *If Davide comes to the party, then, if Carlo doesn't come then Angelo comes*
5. *Carlo comes to the party provided that Davide doesn't come, but, if Davide comes, then Bruno doesn't come*
6. *A necessary condition for Angelo coming to the party, is that, if Bruno and Carlo aren't coming, Davide comes*
7. *Angelo, Bruno and Carlo come to the party if and only if Davide doesn't come, but, if neither Angelo nor Bruno come, then Davide comes only if Carlo comes*

Exercise 1.14 Angelo, Bruno and Carlo are three students that took the Logic exam. Let's consider a propositional language where

- A means *Aldo passed the exam*,
- B means *Bruno passed the exam*,
- C means *Carlo passed the exam*.

Formalize the following sentences:

1. *Carlo is the only one passing the exam*
2. *Aldo is the only one not passing the exam*
3. *Only one, among Aldo, Bruno and Carlo, passed the exam*
4. *At least one among Aldo, Bruno and Carlo passed*
5. *At least two among Aldo, Bruno and Carlo passed the exam*

6. *At most two among Aldo, Bruno and Carlo passed the exam*
7. *Exactly two, among Aldo, Bruno and Carlo passed the exam*

Exercise 1.15 Indicate which single logical connective should be used in the translation into Logical Propositions ($\mathcal{LOP}$) of the sentence "I am going to Rome by train or plane."

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.16 Indicate which single logical connective should be used in the translation into Logical Propositions ($\mathcal{LOP}$) of the sentence "I took the umbrella but got wet."

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.17 Indicate which single logical connective should be used in the translation into Logical Propositions ($\mathcal{LOP}$) of the sentence "Fausto's house and Vincenzo's house are close together."

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.18 Indicate which single logical connective should be used in the translation into Logical Propositions ($\mathcal{LOP}$) of the sentence "Nicole goes to the mountain or to the swimming pool".

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.19 Indicate which single logical connective should be used in the translation into Logical Propositions ($\mathcal{LOP}$) of the sentence "Luigi and Nicole are together in the mountain".

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.20 Indicate which single logical connective should be used in the translation into propositional logic (LOP) of the sentence "If Roberto studies, then he will go out in the evening"

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.21 Indicate which single logical connective should be used in the translation into propositional logic (LOP) of the sentence “Roberto studied more than Chiara”.

1. $\wedge$ (and)
2. $\vee$ (or)
3. $\neg$ (not)
4. $\oplus$ (xor)
5. $\supset$ (implication)
6. $\equiv$ (equivalence)
7. No logical connective

Exercise 1.22 Considering the following mapping from informal text to symbols:

- O: “I’ll take my umbrella.”
- P: “It’s raining today.”
- E: “I’m going out today.”
- D: “I’ll go out tomorrow.”

Indicate which of the following statements are true:

1. The formalization in LoP of “Tomorrow I will go out only if it rains today and I don’t go out” is $(P \wedge \neg E) \supset D$
2. The formalization in LoP of “Today I will go out even if it rains” is $E \wedge (P \vee \neg P)$
3. The formalization in LoP of “The fact that I take an umbrella is a necessary condition for the fact that it is raining today and that I am going out” is $\neg O \supset (\neg P \vee \neg E)$
4. The formalization in LoP of “Tomorrow I will go out unless I go out today” is $D \oplus E$

1.4 Normal forms

Exercise 1.23 Convert the following formula from LoP to CNF, showing the intermediate steps of the procedure.

$$\neg((a \supset (b \vee c)) \supset ((b \wedge a) \supset \neg(c \vee b)))$$

Exercise 1.24 Convert the following formula from LoP to CNF, showing the intermediate steps of the procedure.

$$((a \wedge b) \supset \neg(c \vee a)) \wedge (c \supset (\neg b \wedge a))$$

Exercise 1.25 Convert the following formula from LoP to CNF, showing the intermediate steps of the procedure.

$$((a \vee b) \supset \neg(b \wedge c)) \supset (\neg(b \wedge \neg d) \vee (\neg a \wedge d))$$

Exercise 1.26 Indicate which of the following are well-formed formulas in CNF.

1. $(a \wedge b) \vee (b \wedge c) \vee (\neg d \wedge a) \vee \neg c$
2. $(a \wedge b) \vee \neg(b \wedge c) \vee (\neg d \wedge a) \vee c$
3. $\neg c$
4. $a \vee b \vee c \vee d \vee \neg c$
5. $(a \wedge b \wedge c \wedge d \wedge e) \vee (\neg a \wedge b \wedge c \wedge d \wedge f) \vee (a \wedge b \wedge c \wedge d \wedge a)$
6. $a \wedge b \wedge c \wedge d \wedge e \wedge f$

Exercise 1.27 Indicate which of the following are well-formed formulas in CNF.

1. $a \wedge (b \vee c) \wedge (\neg d \vee a \vee c) \wedge \neg c$

2. $b \wedge \neg(b \vee \neg c \vee a) \wedge (\neg d \vee a)$
3. $\neg a$
4. $a \vee b \vee \neg c$
5. $(a \wedge b \wedge c) \vee (\neg a \wedge b \wedge c) \vee (a \wedge d \wedge a)$
6. $a \wedge b \wedge c \wedge d \wedge e \wedge f$
7. $(a \vee b) \wedge (c \wedge b)$

Exercise 1.28 Indicate which of the following are well-formed formulas in DNF

1. $(a \wedge b) \vee (b \wedge c) \vee (d \wedge \neg e)$
2. $(a \wedge b \wedge d \wedge \neg e) \vee (a \vee b \wedge e)$
3. $(a \wedge c \wedge \neg d) \vee (b \wedge c \supset e)$
4. $a \wedge b \wedge \neg c$
5. a
6. $a \vee b \vee c$
7. $(a \wedge b) \vee \neg(c \wedge a)$

1.5 Reasoning problems

Exercise 1.29 Let's consider the interpretation v where $v(p) = \perp$, $v(q) = \top$, $v(r) = \top$. Does v satisfy the following propositional formulas?

1. $(p \supset \neg q) \vee \neg(r \wedge q)$
2. $(\neg p \vee \neg q) \supset (p \vee \neg r)$
3. $\neg(\neg p \supset \neg q) \wedge r$
4. $\neg(\neg p \supset q \wedge \neg r)$

Exercise 1.30 RECALL: $\Gamma \models \Phi$ iff all models satisfying the formulas in Γ also satisfy Φ .
Let A, B, C be propositions. If $A \models B \wedge C$, then $A \models B$ or $A \models C$ or both?

Exercise 1.31 Let A, B, C, D, E be propositions, and given that:

- $P = (A \vee B) \wedge (\neg C \vee \neg D \vee E)$
- $Q1 = A \vee B$
- $Q2 = (A \vee B \vee C) \wedge ((B \wedge C \wedge D) \supset E)$
- $Q3 = (A \vee B) \wedge (\neg D \vee E)$

Which of the following is true?

1. $P \models Q1$
2. $P \models Q2$
3. $P \models Q3$

Exercise 1.32 Compute the truth table of $(F \vee G) \wedge \neg(F \wedge G)$.

Exercise 1.33 Check if $(P \wedge Q) \vee (R \supset S)$ is satisfiable using the truth tables.

Exercise 1.34 Check if $(P \wedge Q) \vee (R \supset S)$ is valid using the truth tables.

Exercise 1.35 Check if $(P \supset Q) \vee (P \supset \neg Q)$ is valid using the truth tables.

Exercise 1.36 Check if $(\neg P \vee Q) \wedge (Q \supset \neg R \wedge \neg P) \wedge (P \vee R)$ is satisfiable using the truth tables.

Exercise 1.37 Check if $\varphi : P \wedge \neg Q \supset P \wedge Q$ is a logical consequence of the formula $\psi : \neg P$ using the truth tables.

Exercise 1.38 Check if $P \supset (Q \wedge \neg Q)$ and $\neg P$ are logically equivalent using the truth tables.

Exercise 1.39 Check if $\varphi : \neg p$ is a logical consequence of the formula $\psi : ((q \vee r) \wedge q) \supset (\neg p \wedge \neg(r \wedge p))$ using the truth tables.

Exercise 1.40 Check if $\varphi : p \supset (q \wedge \neg q)$ and $\psi : \neg p$ are logically equivalent using the truth tables.

Exercise 1.41 Check if $(A \wedge B) \wedge \neg B$ is unsatisfiable using the truth tables.

Exercise 1.42 Check if $\neg(A \supset B) \supset (A \wedge \neg B)$ is valid using the truth tables.

Exercise 1.43 Compute the truth tables for the following propositional formulas:

1. $(p \supset p) \supset p$
2. $p \supset (p \supset p)$
3. $p \vee q \supset p \wedge q$
4. $p \vee (q \wedge r) \supset (p \wedge r) \vee q$
5. $p \supset (q \supset p)$
6. $(p \wedge \neg q) \vee \neg(p \supset q)$

Exercise 1.44 Use the truth table method to verify whether the following formulas are valid, satisfiable or unsatisfiable:

1. $(p \supset q) \wedge \neg q \supset \neg p$
2. $(p \supset q) \supset (p \supset \neg q)$
3. $(p \vee q \supset r) \vee p \vee q$
4. $(p \vee q) \wedge (p \supset r \wedge q) \wedge (q \supset \neg r \wedge p)$
5. $(p \supset (q \supset r)) \supset ((p \supset q) \supset (p \supset r))$
6. $(p \vee q) \wedge (\neg q \wedge \neg p)$
7. $(\neg p \supset q) \vee ((p \wedge \neg r) \supset q)$
8. $(p \supset q) \wedge (p \supset \neg q)$
9. $(p \supset (q \vee r)) \vee (r \supset \neg p)$

Exercise 1.45 Use the truth table method to verify whether the following logical consequences and equivalences are correct:

1. $(p \supset q) \models \neg p \supset \neg q$
2. $(p \supset q) \wedge \neg q \models \neg p$
3. $p \supset q \wedge r \models (p \supset q) \supset r$
4. $p \vee (\neg q \wedge r) \models q \vee \neg r \supset p$
5. $\neg(p \wedge q) \equiv \neg p \vee \neg q$
6. $(p \vee q) \wedge (\neg p \supset \neg q) \equiv q$
7. $(p \wedge q) \vee r \equiv (p \supset \neg q) \supset r$
8. $(p \vee q) \wedge (\neg p \supset \neg q) \equiv p$
9. $((p \supset q) \supset q) \supset q \equiv p \supset q$

Exercise 1.46 Find an interpretation that does not satisfy the following formula:

$$(\neg(A \vee B) \wedge (C \wedge \neg D)) \supset (\neg C \vee B)$$

specifying for each propositional variable the corresponding truth value assigned by the interpretation (True, False, Unassigned)

Exercise 1.47 List the models for the following formulas using Truth Tables, given the propositions A, B, C :

1. $A \wedge \neg B$
2. $(A \wedge B) \vee (B \wedge C)$
3. $(A \vee B) \supset C$
4. $\neg A \equiv B \equiv C$
5. $(A \vee B) \supset \neg C$

Exercise 1.48 Apply the final version of the DPLL procedure to find as many models as possible.

$$(a \vee \neg b) \wedge (b \vee c \vee \neg a) \wedge (d \vee \neg c) \wedge (\neg d \vee a)$$

Exercise 1.49 Apply the final version of the DPLL procedure to find as many models as possible.

$$(a \vee \neg b \vee \neg c) \wedge (\neg a \vee \neg b \vee d) \wedge (\neg a \vee b \vee \neg d) \wedge \neg c$$

Exercise 1.50 Consider the final version of the DPLL algorithm seen in class, applying the different kind of literal selection with the following priority (highest to lowest): unit-propagation, pure-literal, choice from the leftmost literal. Consider then the following CNF formula, determine which of the statements are true.

$$(a \vee \neg b) \wedge (c \vee d \vee e) \wedge (\neg a \vee \neg d) \wedge (\neg c \vee d \vee \neg f) \wedge (f \vee g)$$

1. $\{\neg a, \neg b, d, e, f, g\}$ is a model of the formula
2. $\{\neg a, \neg b, \neg c, e, f, \neg g\}$ is a model of the formula that can be reached with the DPLL algorithm (considering the constraints in the question).
3. The first step of the DPLL algorithm applied on the formula (considering the constraints in the question) is always a pure-literal assignment.
4. The first step of the DPLL algorithm applied on the formula (considering the constraints in the question) may be a unit-propagation assignment.
5. There isn't a model of the formula that contains both $\neg f$ and $\neg g$
6. $\{\neg a, \neg b, \neg c, e, g\}$ is a model of the formula that can be reached with the DPLL algorithm (considering the constraints in the question).

Exercise 1.51 Determine if the following formula is satisfiable using DPLL:

$$(\neg a \vee c) \wedge (a \vee \neg b \vee c) \wedge (b \vee \neg d) \wedge (\neg c \vee e) \wedge (d \vee c) \wedge \neg e$$

Exercise 1.52 Determine if the following formula is satisfiable using DPLL:

$$(\neg a \vee c) \wedge (a \vee \neg b \vee c) \wedge (b \vee \neg d) \wedge (\neg c \vee \neg d) \wedge (d \vee c)$$

Exercise 1.53 Indicate which of the following statements regarding the following formulas in LOP language are true.

- $A : (a \vee b) \wedge (b \supset \neg(b \wedge a))$
- $B : \neg(b \vee a)$
- $C : b \supset a$

1. C is a logical consequence of $\neg A$
2. $\neg A \supset C$ it is not a tautology.
3. A is a logical consequence of $\neg B$
4. $A \models \neg B$
5. $(A \supset B) \equiv \neg A$
6. $C \vee B$ it is a tautology.

Exercise 1.54 Indicate which of the following statements regarding the following formulas in LOP language are true.

- $A : \neg((\neg a \wedge \neg b) \vee (b \wedge (b \wedge a)))$
- $B : (\neg b \wedge \neg a)$
- $C : \neg b \vee a$

1. C is a logical consequence of $\neg A$
2. $\neg A \supset C$ it is not a tautology.
3. A is a logical consequence of $\neg B$
4. $A \models \neg B$
5. $(A \supset B) \equiv \neg A$
6. $C \vee B$ it is a tautology.

1.6 Solutions

Solution 1.1 $\neg \vee \equiv \supset \wedge \models \oplus$

Solution 1.2 2, 3.

Solution 1.3 2, 5.

Solution **1.4** 2, 5.

Solution **1.5** 1, 2, 6, 8.

Solution **1.6** 1, 5.

Solution **1.7** 2, 6.

Solution **1.8** 2, 3, 4, 6, 8.

Solution **1.9**

1. True, as per the definition of $\mathcal{LOP}$ interpretation function
2. True, as per the definition of $\mathcal{LOP}$ interpretation function
3. False, as per the definition of interpretation function of $\mathcal{LOP}$, the translation function is one-to-one.
4. False, in $\mathcal{LODE}$ for each theory there is one and only one model that encodes its intended meaning.

Solution **1.10** We have the following:

1. $p \wedge q \supset \neg r$
2. $p \supset q$
3. $\neg(p \wedge \neg q)$..which is equivalent to $p \supset q$

(NOTE) Notice how the precision of formal languages avoid the ambiguities of natural languages.

Solution **1.11** For both statements 1. and 2. we have $p \supset q$.

Solution **1.12** We have the following.

1. $\neg A$
2. $A \wedge B$
3. $A \supset \neg B$
4. $A \vee (\neg A \supset B)$ is logically equivalent to $A \vee B$
5. $(A \wedge B) \vee (\neg A \wedge \neg B)$ is logically equivalent to $A \equiv B$

Solution **1.13** We have the following

1. $D \supset B \wedge C$
2. $C \supset \neg A \wedge \neg B$
3. $D \equiv (C \wedge \neg A)$
4. $D \supset (\neg C \supset A)$
5. $(\neg D \supset C) \wedge (D \supset \neg B)$
6. $A \supset (\neg B \wedge \neg C \supset D)$
7. $(A \wedge B \wedge C \equiv \neg D) \wedge (\neg A \wedge \neg B \supset (D \equiv C))$

Solution **1.14**

1. $C \wedge \neg A \wedge \neg B$
2. $C \wedge \neg A \wedge B$
3. $A \oplus B \oplus C$ or $(A \wedge \neg B \wedge \neg C) \vee (\neg A \wedge B \wedge \neg C) \vee (\neg A \wedge \neg B \wedge C)$
4. $A \vee B \vee C$
5. $(A \wedge B) \vee (B \wedge C) \vee (A \wedge C)$
6. $\neg(A \wedge B \wedge C)$
7. $(A \wedge B) \oplus (B \wedge C) \oplus (A \wedge C)$

Solution **1.15** 4.

Solution **1.16** 1.

Solution **1.17** 7.

Solution **1.18** 2.

Solution **1.19** 1.

Solution **1.20** 5.

Solution **1.21** 7.

Solution 1.22

1. False, the correct version is $D \supset (P \wedge \neg E)$
2. True
3. True
4. True

Solution 1.23

$$\begin{aligned}
& \text{CNF}(\neg((a \supset (b \vee c)) \supset ((b \wedge a) \supset \neg(c \vee b)))) \\
& \text{CNF}(a \supset (b \vee c)) \wedge \text{CNF}(\neg((b \wedge a) \supset \neg(c \vee b))) \\
& (\text{CNF}(\neg a) \otimes \text{CNF}(b \vee c)) \wedge (\text{CNF}(b \wedge a) \wedge \text{CNF}(c \vee b)) \\
& (\text{CNF}(\neg a) \otimes (\text{CNF}(b) \otimes \text{CNF}(c))) \wedge (((\text{CNF}(b) \wedge \text{CNF}(a)) \wedge (\text{CNF}(c) \otimes \text{CNF}(b))) \\
& (\neg a \otimes (b \otimes c)) \wedge ((b \wedge a) \wedge (c \otimes b)) \\
& (\neg a \otimes (b \vee c)) \wedge b \wedge a \wedge (c \vee b) \\
& ((\neg a \vee b \vee c)) \wedge b \wedge a \wedge (c \vee b) \\
& a \wedge b \wedge (b \vee c) \wedge (\neg a \vee b \vee c)
\end{aligned}$$

Solution 1.24

$$\begin{aligned}
& ((a \wedge b) \supset \neg(c \vee a)) \wedge (c \supset (\neg b \wedge a)) \\
& \text{CNF}(((a \wedge b) \supset \neg(c \vee a)) \wedge (c \supset (\neg b \wedge a))) \\
& \text{CNF}(((a \wedge b) \supset \neg(c \vee a))) \wedge \text{CNF}((c \supset (\neg b \wedge a))) \\
& (\text{CNF}(\neg(a \wedge b)) \otimes \text{CNF}(\neg(c \vee a))) \wedge (\text{CNF}(\neg c) \otimes \text{CNF}((\neg b \wedge a))) \\
& ((\text{CNF}(\neg a) \otimes \text{CNF}(\neg b)) \otimes (\text{CNF}(\neg c) \wedge \text{CNF}(\neg a))) \wedge (\text{CNF}(\neg c) \otimes (\text{CNF}(\neg b) \wedge \text{CNF}(a))) \\
& ((\neg a \otimes \neg b) \otimes (\neg c \wedge \neg a)) \wedge (\neg c \otimes (\neg b \wedge a)) \\
& ((\neg a \vee \neg b) \otimes (\neg c \wedge \neg a)) \wedge (\neg c \otimes (\neg b \wedge a)) \\
& ((\neg a \vee \neg b \vee \neg c) \wedge (\neg a \vee \neg b \vee \neg a)) \wedge (\neg c \otimes (\neg b \wedge a)) \\
& ((\neg a \vee \neg b \vee \neg c) \wedge (\neg a \vee \neg b \vee \neg a)) \wedge ((\neg c \vee \neg b) \wedge (\neg c \vee a)) \\
& (\neg a \vee \neg b \vee \neg c) \wedge (\neg b \vee \neg a) \wedge (\neg c \vee \neg b) \wedge (\neg c \vee a) \\
& (a \vee \neg c) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg b) \wedge (\neg a \vee \neg b \vee \neg c)
\end{aligned}$$

Solution 1.25

$$\begin{aligned}
& \text{CNF}(((a \vee b) \supset \neg(b \wedge c)) \supset (\neg(b \wedge \neg d) \vee (\neg a \wedge d))) \\
& \text{CNF}(\neg((a \vee b) \supset \neg(b \wedge c)) \otimes \text{CNF}(\neg(b \wedge \neg d) \vee (\neg a \wedge d))) \\
& (\text{CNF}((a \vee b)) \wedge \text{CNF}((b \wedge c))) \otimes (\text{CNF}(\neg(b \wedge \neg d)) \otimes \text{CNF}((\neg a \wedge d))) \\
& ((\text{CNF}(a) \otimes \text{CNF}(b)) \wedge (\text{CNF}(b) \wedge \text{CNF}(c))) \otimes ((\text{CNF}(\neg b) \otimes \text{CNF}(d)) \otimes (\text{CNF}(\neg a) \wedge \text{CNF}(d))) \\
& ((a \otimes b) \wedge (b \wedge c)) \otimes ((\neg b \otimes d) \otimes (\neg a \wedge d)) \\
& ((a \vee b) \wedge b \wedge c) \otimes ((\neg b \vee d) \otimes (\neg a \wedge d)) \\
& ((a \vee b) \wedge b \wedge c) \otimes ((\neg b \vee d \vee \neg a) \wedge (\neg b \vee d \vee d)) \\
& (a \vee b \vee \neg b \vee d \vee \neg a) \wedge (a \vee b \vee \neg b \vee d \vee d) \wedge \dots \\
& \dots (b \vee \neg b \vee d \vee \neg a) \wedge (b \vee \neg b \vee d \vee d) \wedge (c \vee \neg b \vee d \vee \neg a) \wedge (c \vee \neg b \vee d \vee d) \\
& (c \vee \neg b \vee d \vee \neg a) \wedge (c \vee \neg b \vee d)
\end{aligned}$$

Solution 1.26

1. False
2. False
3. True
4. True

5. False
6. True

Solution 1.27

1. True
2. False
3. True
4. True
5. False
6. True
7. False

Solution 1.28

1. True
2. False
3. False
4. True
5. True
6. True
7. False

Solution 1.29 v satisfies formulas 1., 3. and 4. v doesn't satisfy 2.

Solution 1.30 PROOF:

If $A \models B \wedge C$, for all I such that $I(A) = \text{True}$ it should be $I(B \wedge C) = \text{True}$.

However, by definition this means that $I(B) = \text{True}$ and $I(C) = \text{True}$.

Therefore both $A \models B$ and $A \models C$.

Solution 1.31

1. True
2. True
3. False

Solution 1.32

F	G	$F \vee G$	$F \wedge G$	$\neg(F \wedge G)$	$(F \vee G) \wedge \neg(F \wedge G)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Solution 1.33 $(P \wedge Q) \vee (R \supset S)$ is satisfiable.

P	Q	R	S	$P \wedge Q$	$R \supset S$	$(P \wedge Q) \vee (R \supset S)$
T	T	T	T	T	T	T
T	T	T	F	T	F	T
T	T	F	T	T	T	T
T	F	T	T	F	T	T
F	T	T	T	F	T	T
T	T	F	F	T	T	T
T	F	T	F	F	F	F
F	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	F	T	T
F	F	F	T	F	T	T
F	F	F	F	F	T	T

Solution 1.34 $(P \wedge Q) \vee (R \supset S)$ is not valid, check the truth table in the previous exercise.

Solution 1.35 $(P \supset Q) \vee (P \supset \neg Q)$ is valid.

P	Q	$P \supset Q$	$P \supset \neg Q$	$(P \supset Q) \vee (P \supset \neg Q)$
T	T	T	F	T
T	F	F	T	T
F	T	T	T	T
F	F	T	T	T

Solution **1.36** $(\neg p \vee q) \wedge (q \supset \neg r \wedge \neg p) \wedge (p \vee r)$ is satisfiable.

Let $\varphi : (\neg p \vee q) \wedge (q \supset \neg r \wedge \neg p) \wedge (p \vee r)$

p	q	r	$\neg p \vee q$	$\neg r \wedge \neg p$	$q \supset \neg r \wedge \neg p$	$(p \vee r)$	φ
T	T	T	T	F	F	T	F
T	T	F	T	F	F	T	F
T	F	T	F	F	T	T	F
T	F	F	F	F	T	T	F
F	T	T	T	F	F	T	F
F	T	F	T	T	T	F	F
F	F	T	T	F	T	T	T
F	F	F	T	T	T	F	F

Solution **1.37** $\varphi : p \wedge \neg q \supset p \wedge q$ is a logical consequence of the formula $\psi : \neg p$, i.e. $\psi \models \varphi$.

p	q	$\psi : \neg p$	$p \wedge \neg q$	$p \wedge q$	$\varphi : p \wedge \neg q \supset p \wedge q$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	F	F	T

Solution **1.38** $p \supset (q \wedge \neg q)$ and $\neg p$ are logically equivalent.

p	q	$q \wedge \neg q$	$p \supset (q \wedge \neg q)$	$\neg p$
T	T	F	F	F
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T

Solution **1.39** $\varphi : \neg p$ is not a logical consequence of the formula $\psi : ((q \vee r) \wedge q) \supset (\neg p \wedge \neg(r \wedge p))$.

p	q	r	$\psi_1 : q \vee r$	$\psi_1 \wedge q$	$\psi_2 : r \wedge p$	$\neg p \wedge \neg \psi_2$	ψ	φ
T	T	T	T	T	T	F	F	F
T	T	F	T	T	F	F	F	F
T	F	T	T	F	T	F	F	F
T	F	F	F	F	F	F	T	F
F	T	T	T	T	F	F	F	T
F	T	F	T	T	F	F	F	T
F	F	T	T	F	F	F	F	T
F	F	F	T	F	F	F	F	T

Solution **1.40** $\varphi : p \supset (q \wedge \neg q)$ and $\psi : \neg p$ are logically equivalent.

p	q	$q \wedge \neg q$	$\varphi : p \supset (q \wedge \neg q)$	$\psi : \neg p$
T	T	F	F	F
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T

Solution **1.41** $(A \wedge B) \wedge \neg B$ is unsatisfiable.

A	B	$A \wedge B$	$\neg B$	$(A \wedge B) \wedge \neg B$
T	T	T	F	F
T	F	F	T	F
F	T	F	F	F
F	F	F	T	F

Solution **1.42** Same procedure as the other exercises.

Solution **1.43** Same procedure as the other exercises.

Solution **1.44** Same procedure as the other exercises.

Solution **1.45** Same procedure as the other exercises.

Solution **1.46** Same procedure as the other exercises.

Solution **1.47** Same procedure as the other exercises.

Solution **1.48** Models for $(a \vee \neg b) \wedge (b \vee c \vee \neg a) \wedge (d \vee \neg c) \wedge (\neg d \vee a)$:

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
 2. $\{\{\top, \neg b\}, \{b, c, \perp\}, \{d, \neg c\}, \{\neg d, \top\}\}|\{a\}$ select a
 3. $\{\{b, c\}, \{d, \neg c\}\}|\{a\}$
 4. $\{\{\top, c\}, \{d, \neg c\}\}|\{a, b\}$ pure-literal b
 5. $\{\{d, \neg c\}\}|\{a, b\}$
 6. $\{\{\top, \neg c\}\}|\{a, b, d\}$ pure-literal d
 7. $\{\}|\{a, b, d\}$ $\{a, b, d\}$ is a model, so also $\{a, b, d, c\}$ and $\{a, b, d, \neg c\}$ are models
-
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
 2. $\{\{\top, \neg b\}, \{b, c, \perp\}, \{d, \neg c\}, \{\neg d, \top\}\}|\{a\}$ select a
 3. $\{\{b, c\}, \{d, \neg c\}\}|\{a\}$
 4. $\{\{b, c\}, \{\top, \neg c\}\}|\{a, d\}$ pure-literal d
 5. $\{\{b, c\}\}|\{a, d\}$
 6. From here applying pure-literal to b will result in the same model as before, applying pure literal to c we can see that $\{a, d, c\}$ is a model so also $\{a, d, c, \neg b\}$ is a model, we already know that $\{a, d, c, b\}$ is a model.
-
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
 2. $\{\{a, \perp\}, \{\top, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{b\}$ select b
 3. $\{\{a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{b\}$
 4. $\{\{\top\}, \{d, \neg c\}, \{\neg d, \top\}\}|\{b, a\}$ unit-propagation a
 5. $\{\{d, \neg c\}\}|\{b, a\}$
 6. From here if we apply pure-literal to d we get a model that we already discovered, so we apply pure-literal to $\neg c$ and see that $\{b, a, \neg c\}$ is a model so also $\{b, a, \neg c, \neg d\}$ is a model we already know that $\{b, a, \neg c, d\}$ is a model.
-
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
 2. $\{\{a, \neg b\}, \{b, \top, \neg a\}, \{d, \perp\}, \{\neg d, a\}\}|\{c\}$ select c
 3. $\{\{a, \neg b\}, \{d\}, \{\neg d, a\}\}|\{c\}$
 4. $\{\{a, \neg b\}, \{\top\}, \{\perp, a\}\}|\{c, d\}$ unit-propagation d
 5. $\{\{a, \neg b\}, \{a\}\}|\{c, d\}$
 6. $\{\{\top, \neg b\}, \{\top\}\}|\{c, d, a\}$ unit-propagation a
 7. $\{\}|\{c, d, a\}$ we had already found $\{c, d, a\}$ as a model.
-
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
 2. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{\top, \neg c\}, \{\perp, a\}\}|\{d\}$ select d
 3. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{a\}\}|\{d\}$
 4. $\{\{\top, \neg b\}, \{b, c, \perp\}, \{\top\}\}|\{d, a\}$ unit-propagation a
 5. $\{\{b, c\}\}|\{d, a\}$
 6. From here we can either chose to apply unit-propagation to b or c and we will find models that we have already found.
-
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
 2. $\{\{\perp, \neg b\}, \{b, c, \top\}, \{d, \neg c\}, \{\neg d, \perp\}\}|\{\neg a\}$ select $\neg a$
 3. $\{\{\neg b\}, \{d, \neg c\}, \{\neg d\}\}|\{\neg a\}$
 4. $\{\{\top\}, \{d, \neg c\}, \{\neg d\}\}|\{\neg a, \neg b\}$ unit-propagation $\neg b$
 5. $\{\{d, \neg c\}, \{\neg d\}\}|\{\neg a, \neg b\}$
 6. $\{\{\perp, \neg c\}, \{\top\}\}|\{\neg a, \neg b, \neg d\}$ unit-propagation $\neg d$
 7. $\{\{\neg c\}\}|\{\neg a, \neg b, \neg d\}$

8. $\{\{\top\}\}|\{\neg a, \neg b, \neg d, \neg c\}$ unit-propagation $\neg c$
9. $\{\}\mid\{\neg a, \neg b, \neg d, \neg c\}$ **$\{\neg a, \neg b, \neg d, \neg c\}$ is a model.**
10. We can see that if at step 4 we had chosen to apply unit-propagation to $\neg d$, the result would be the same.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \top\}, \{\perp, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg b$
3. $\{c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$
4. $\{\top, \neg a\}, \{d, \perp\}, \{\neg d, a\}\}|\{\neg b, c\}$ select c
5. $\{d\}, \{\neg d, a\}\}|\{\neg b, c\}$
6. $\{\top\}, \{\perp, a\}\}|\{\neg b, c, d\}$ unit-propagation d
7. $\{a\}\}|\{\neg b, c, d\}$
8. $\{\top\}\}|\{\neg b, c, d, a\}$ unit-propagation a
9. $\{\}\}|\{\neg b, c, d, a\}$ $\{\neg b, c, d, a\}$ is a model, which we already know.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \top\}, \{\perp, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg b$
3. $\{c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$
4. $\{\perp, \neg a\}, \{d, \top\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg c$
5. $\{\neg a\}, \{\neg d, a\}\}|\{\neg b, \neg c\}$
6. $\{\top\}, \{\neg d, \perp\}\}|\{\neg b, \neg c\}$ unit-propagation $\neg a$
7. $\{\{\neg d\}\}|\{\neg b, \neg c, \neg a\}$
8. From here we can see that we would find a model that we have already found.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \top\}, \{\perp, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg b$
3. $\{c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$
4. $\{c, \perp\}, \{d, \neg c\}, \{\neg d, \top\}\}|\{\neg b, a\}$ select a
5. $\{c\}, \{d, \neg c\}\}|\{\neg b, a\}$
6. From here, by applying unit-propagation to c we see that we find a model that we have already found.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \top\}, \{\perp, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg b$
3. $\{c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$
4. $\{c, \top\}, \{d, \neg c\}, \{\neg d, \perp\}\}|\{\neg b, \neg a\}$ select $\neg a$
5. $\{d, \neg c\}, \{\neg d\}\}|\{\neg b, \neg a\}$
6. From here, by applying unit-propagation to $\neg d$ we see that we find a model that we have already found.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \top\}, \{\perp, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg b$
3. $\{c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$
4. $\{c, \neg a\}, \{\top, \neg c\}, \{\perp, a\}\}|\{\neg b, d\}$ select d
5. $\{c, \neg a\}, \{a\}\}|\{\neg b, d\}$
6. Again, by propagating a we find an already found model.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \top\}, \{\perp, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$ select $\neg b$
3. $\{c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\neg b\}$
4. $\{c, \neg a\}, \{\perp, \neg c\}, \{\top, a\}\}|\{\neg b, \neg d\}$ select $\neg d$
5. $\{c, \neg a\}, \{\neg c\}\}|\{\neg b, \neg d\}$
6. Again, by propagating $\neg c$ we find an already found model.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\}|\{\}$
2. $\{\{a, \neg b\}, \{b, \perp, \neg a\}, \{d, \top\}, \{\neg d, a\}\}|\{\neg c\}$ select $\neg c$
3. $\{\{a, \neg b\}, \{b, \neg a\}, \{\neg d, a\}\}|\{\neg c\}$
4. $\{\{a, \neg b\}, \{b, \neg a\}, \{\top, a\}\}|\{\neg c, \neg d\}$ pure-literal $\neg d$
5. $\{\{a, \neg b\}, \{b, \neg a\}\}|\{\neg c, \neg d\}$
6. From here it's clear that by choosing a or b we get the model $\{\neg c, \neg d, a, b\}$.
7. Instead if we choose $\neg a$ or $\neg b$ we get the model $\{\neg c, \neg d, \neg a, \neg b\}$ again.

1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\} \mid \{\}$
2. $\{\{a, \neg b\}, \{b, \top, \neg a\}, \{d, \perp\}, \{\neg d, a\}\} \mid \{c\}$ select c
3. $\{\{a, \neg b\}, \{d\}, \{\neg d, a\}\} \mid \{c\}$
4. $\{\{a, \neg b\}, \{\top\}, \{\perp, a\}\} \mid \{c, d\}$ unit-propagation d
5. $\{\{a, \neg b\}, \{a\}\} \mid \{c, d\}$
6. By propagating a we can see that we would end on the model $\{a, d, c, \neg b\}$.
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\} \mid \{\}$
2. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{\top, \neg c\}, \{\perp, a\}\} \mid \{d\}$ select d
3. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{a\}\} \mid \{d\}$
4. $\{\{\top, \neg b\}, \{b, c, \perp\}, \{\top\}\} \mid \{d, a\}$ unit-propagation a
5. $\{\{b, c\}\} \mid \{d, a\}$
6. No matter the choice here, we will find a model that we have already found.
1. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{d, \neg c\}, \{\neg d, a\}\} \mid \{\}$
2. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{\perp, \neg c\}, \{\top, a\}\} \mid \{\neg d\}$ select $\neg d$
3. $\{\{a, \neg b\}, \{b, c, \neg a\}, \{\neg c\}\} \mid \{\neg d\}$
4. $\{\{a, \neg b\}, \{b, \perp, \neg a\}, \{\top\}\} \mid \{\neg d, \neg c\}$ unit-propagation $\neg c$
5. $\{\{a, \neg b\}, \{b, \neg a\}\} \mid \{\neg d, \neg c\}$
6. Here we are in a situation that we have already seen, therefore there are no more models.

Solution **1.49** Apply the same procedure as the exercise before.

Solution **1.50**

1. False
2. False
3. True
4. False
5. True
6. True

Solution **1.51** Unsatisfiable.

Solution **1.52** Satisfiable with $\{\neg a, \neg b, \neg d, c\}$.

Solution **1.53**

1. True
2. False
3. False
4. True
5. True
6. False

a	b	A	B	C	$\neg A \supset C$	$(A \supset B)$	$\neg A$	$\neg B$	$C \vee B$
T	T	F	F	T	T	T	T	T	T
T	F	T	F	T	T	F	F	T	T
F	T	T	F	F	T	F	F	T	F
F	F	F	T	T	T	T	T	F	T

Solution **1.54**

1. True
2. False
3. False
4. True
5. True
6. False

<i>a</i>	<i>b</i>	<i>A</i>	<i>B</i>	<i>C</i>	$\neg A \supset C$	$(A \supset B)$	$\neg A$	$\neg B$	$C \vee B$
T	T	F	F	T	T	T	T	T	T
T	F	T	F	T	T	F	F	T	T
F	T	T	F	F	T	F	F	T	F
F	F	F	T	T	T	T	T	F	T